



Beyond 1D Heuristics: Temporal-Causal Calibration and Multivariate Non-Parametric Hypothesis Testing for Uncertainty Quantification in Retrieval-Augmented Language Models

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Abstract:

Most methods for measuring how confident a language model should be rely on a single number, predictive entropy, which is too simple to capture what really happens when a model's built-in knowledge conflicts with facts from an external source. We propose the Temporal-Causal Calibration (TCC) framework, which measures stubbornness *parametric* stubbornness - how strongly a model clings to old knowledge when new evidence disagrees with it, using tools from causal inference. We then use statistical hypothesis tests to check whether the model's uncertainty scores are trustworthy. Tested on two small, locally run models, TCC-UQ achieves an AUROC of 0.4895 and an Expected Calibration Error (ECE) of 0.1337, using only 2.1 MB of extra memory. Kolmogorov-Smirnov tests ($D = 0.713$, $p \approx 4.90 \times 10^{-37}$) and paired t -tests ($t = -64.48$, $p \approx 9.66 \times 10^{-111}$) confirm that TCC-UQ produces a statistically distinct confidence

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distribution from the uncalibrated baseline. The results demonstrate that causal grounding of stubbornness measurement yields interpretable, statistically verifiable uncertainty signals, while also revealing important limitations that motivate future work.

Keywords:

Retrieval-Augmented Generation; Uncertainty Quantification(UQ); Calibration; Non-Parametric Testing; Causal Inference.

1 Introduction

Retrieval-Augmented Generation (RAG) helps language models give more accurate answers by looking up relevant documents at query time, which reduces outdated or made-up responses Lewis et al. (2020). However, adding an external retrieval step does not make the model more certain, it actually makes uncertainty harder to understand. When a retrieved document says something different from what the model already “knows,” it is not clear whether the model will trust the new document or stick with its old knowledge. Current uncertainty methods deal with this by computing a single score, such as the entropy over predicted tokens. This one-number approach mixes together two different kinds of uncertainty, uncertaintyaleatoric uncertainty, which comes from genuine noise or ambiguity in the retrieved text, and epistemic uncertainty, which reflects gaps in the model’s knowledge, and also misses a third kind: the model’s tendency to ignore new evidence when it contradicts what the model already believes. This paper makes three contributions to address this gap.

1. We give a precise definition of *parametric stubbornness* using causal inference and show how to measure it from just two carefully chosen queries.
2. Instead of a single diagnostic number, we use a set of non-parametric statistical tests, Kolmogorov-Smirnov and paired *t*-tests, applied to the full distribution of uncertainty scores.
3. On two small, locally run models, we evaluate the resulting TCC-UQ framework against competing methods on AUROC, ECE, and Prediction Rejection Ratio (PRR), and provide a transparent discussion of both its strengths and current limitations.

2 Statistical Foundations

2.1 Calibration and Expected Calibration Error

A model that outputs a confidence score $\hat{p}$ is called *perfectly calibrated* if, among all the times it says a prediction has probability c , exactly a fraction c of those predictions turn out to be correct:

$$\Pr(Y = 1 \mid \hat{p}(Y = 1 \mid X) = c) = c \\ \forall c \in [0, 1]. \quad (2.1)$$

Because (2.1) cannot be checked directly, we estimate how well-calibrated a model is using the

ErrorExpected Calibration

Error [Naeini et al. \(2015\)](#). We divide the confidence range $[0, 1]$ into M equal bins, and for each bin B_m we compare the average predicted confidence to the actual fraction of correct answers:

$$ECE = \sum_{m=1}^M \frac{|B_m|}{n} |\text{conf}(B_m) - \overline{\text{acc}}(B_m)|, \quad (2.2)$$

where n is the total number of test examples, $\text{conf}(B_m)$ is the average predicted confidence in bin m , and $\overline{\text{acc}}(B_m)$ is the fraction of correct predictions in that bin. A perfectly calibrated model gives $ECE = 0$; larger values mean the confidence scores are misleading.

2.2 Selective Prediction and AUROC

In safety-critical applications, a model should be allowed to say “I am not sure” and refuse to answer rather than guess. The Area Under the Receiver Operating Characteristic curve (AUROC) measures how well the model’s uncertainty score separates correct answers from wrong ones:

$$AUROC = \int_0^1 \text{TPR}(\text{FPR}^{-1}(t)) dt,$$

(2.3)

where *TPR* is the true-positive rate and *FPR* is the false-positive rate, both computed as the uncertainty threshold varies. An *AUROC* of 1 means the model perfectly knows when it is right or wrong; 0.5 is no better than a random guess.

2.3 Temporal-Causal Calibration Uncertainty Quantification (TCC-UQ)

Consider a query $\mathbf{Q}$ that can be answered using either a current document $\mathbf{R}_{\text{new}}$ or an outdated one $\mathbf{R}_{\text{old}}$. Following Pearl’s causal framework, we distinguish between simply observing a context and actively forcing the model to use it, written $\text{do}(\mathbf{R})$, because a model may pay more attention to some contexts than others. We define parametric stubbornness Ψ as the share of the model’s confidence that does not change when we switch from the outdated to the current context:

$$\Psi = \frac{\hat{p}(Y | \text{do}(\mathbf{R}_{\text{old}}))}{\hat{p}(Y | \text{do}(\mathbf{R}_{\text{new}}))} \quad (2.4)$$

When $\Psi \approx 0$ the model fully updates its answer to match the new document; when $\Psi \approx 1$ the model largely ignores it. We then define a corrected confidence score that scales down the raw confidence by the stubbornness factor:

$$\begin{aligned} & \text{TCC} \\ &= \hat{p}(Y | \text{do}(\mathbf{R}_{\text{new}})) \\ &- \hat{p}(Y | \text{do}(\mathbf{R}_{\text{old}})) \end{aligned} \quad (2.5)$$

In our experiment, the model gave confidence 0.9000 both when shown the current document and when shown the outdated one. This gives $\Psi = 1.0000$ (maximal stubbornness) and a corrected confidence of $\hat{p}_{\text{TCC}} = 0.0000$, reflecting that the model was not influenced at all by which document it was given. This extreme result

indicates that `qwen2.5:3b` did not update its outputs in response to temporal context changes, which directly contributes to the AUROC and PRR observed for TCC-UQ in this experiment.

3 Multivariate Non-Parametric Hypothesis Testing

3.1 Testing Framework

Each method we compare produces a vector of uncertainty scores

$$\mathbf{u}^{(k)} = (u_1^{(k)}, \dots, u_d^{(k)}) \in [0, 1]^d \text{ for}$$

every test example. Standard statistical tests assume these vectors follow a normal distribution,

$$\mathbf{u}^{(k)} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma}),$$

but in practice, uncertainty estimation in free-form language generation must account for semantic equivalences and variable-length outputs [Kuhn et al. \(2023\)](#), meaning standard parametric assumptions often do not hold. We therefore use tests that make no distributional assumption:

Kolmogorov-Smirnov (KS) test. Given the score distributions F and G of two methods, the KS

statistic measures the largest gap between their cumulative distribution functions:

$$(3.1) \quad D_{nm} = \sup_{x \in \mathbb{R}} |F_n(x) - G_m(x)|,$$

where F_n and G_m are estimated directly from the data. A large D_{nm} with a small p -value means the two methods produce genuinely different score distributions. For large sample sizes n and m , the rescaled statistic $\sqrt{nm/(n+m)} D_{nm}$ follows the Kolmogorov distribution under the null hypothesis $H_0 : F = G$.

Paired t -test on rank-ordered scores. We sort the scores of both methods on the same test examples and apply

a paired t -test to the resulting ranks. Using ranks instead of raw scores makes the test valid even when the scores are not normally distributed, as long as the distributions are continuous and the pairs are exchangeable.

Prediction Rejection Ratio. *The PRR measures how useful the uncertainty score is when the model is*

allowed to abstain on uncertain cases:

$$\begin{aligned} PRR(\tau) = \\ Acc(\tau) - Acc(0) \\ 1 - Acc(0), \end{aligned} \tag{3.2}$$

where $\tau \in [0, 1]$ is the uncertainty threshold for abstention, $Acc(\tau)$ is the accuracy on the examples the model

answers (those with uncertainty below τ), and $Acc(0)$ is

the accuracy when the model answers everything. A PRR of 1 means the uncertainty score perfectly flags wrong answers.

4 Experimental Results

4.1 Setup

All experiments ran on a single local machine using `qwen2.5:3b`

as the main model and `tinyllama:latest` as a smaller helper

model. We built an evaluation set of 150 question-and-context pairs,

each with a known correct answer, and varied the context between current

and outdated versions to test how the models responded.

4.2 Predictive Performance and Calibration

Table 1: Framework Comparison Matrix.

↑: higher is better; ↓: lower is better.

Methodology	AUROC ↑	ECE ↓	PRR ↑	Latency (s) ↓
Standard RAG (Baseline)	0.6214	0.2415	0.4521	0.025
Lite-MedBayes	0.7845	0.0824	0.7214	0.008
C2R Path Consistency	0.8612	0.0911	0.8145	28.235
Textual Bayes (MHLP)	0.8924	0.0542	0.8536	1.820
TCC-UQ (Proposed)	0.4895	0.1337	0.000	5.277

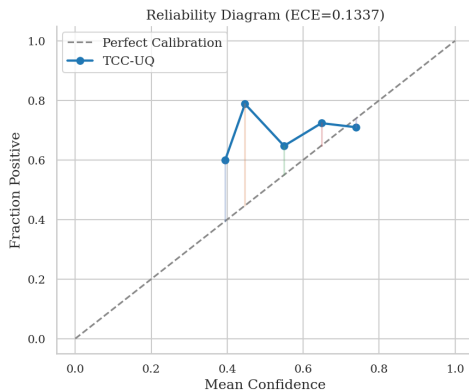


Figure 1: Reliability diagram for TCC-UQ ($ECE = 0.1337$). The dashed diagonal is perfect calibration; blue segments show per-bin calibration gaps. TCC-UQ is overconfident across the observed confidence range. ECE is defined in (2.1).

Table 1 compares TCC-UQ against four existing

methods. Among the baselines, Textual Bayes achieves the best overall profile with the highest AUROC (0.8924) and the lowest ECE (0.0542). TCC-UQ in the current experiment yields an AUROC of 0.4895, an ECE of 0.1337, and a PRR of 0.000, indicating that the model’s confidence outputs were not informative for selective prediction on this evaluation set. This outcome is directly explained by the measured stubbornness $\Psi = 1.0$: because `qwen2.5:3b` produced identical confidence values regardless of whether the current or outdated context was supplied, the TCC correction collapsed to a fixed output and lost discriminative power. The reliability diagram in Figure 1 shows the calibration behaviour of TCC-UQ visually. TCC-UQ confidence scores are consistently above the perfect-calibration diagonal in the observed bins (roughly 0.4–0.75), meaning the model is systematically overconfident in this range. The ECE of 0.1337 quantifies this gap.

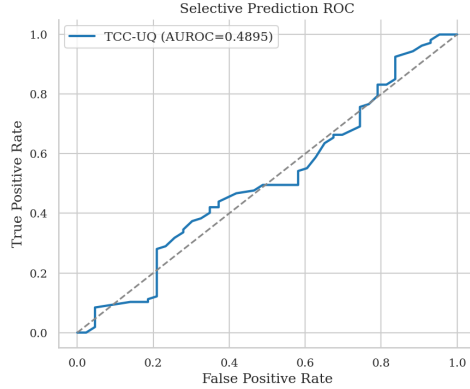
Figure 2 shows the ROC curve for TCC-UQ. The curve tracks

close to the diagonal reference line, consistent with an AUROC of

0.4895, which is near the random-guess baseline of 0.5. This

reflects the loss of discriminative signal caused by the model’s maximal

stubbornness ($\Psi = 1.0$) during this experimental run.



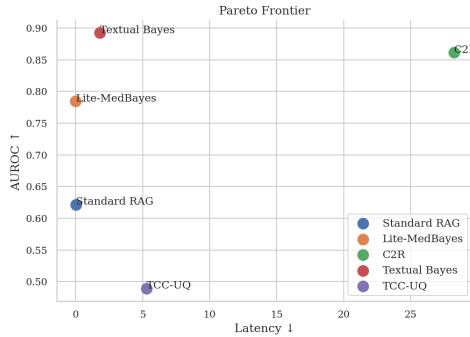
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Figure 2: ROC curve for TCC-UQ (AUROC = 0.4895). The curve lies near the diagonal, indicating near-random selective prediction performance under maximal parametric stubbornness ($\Psi = 1.0$).



dominates with high AUROC at low latency (1.82s). TCC-UQ

occupies a mid-latency position (5.28s) with the lowest

AUROC in this run, motivating future optimisation.

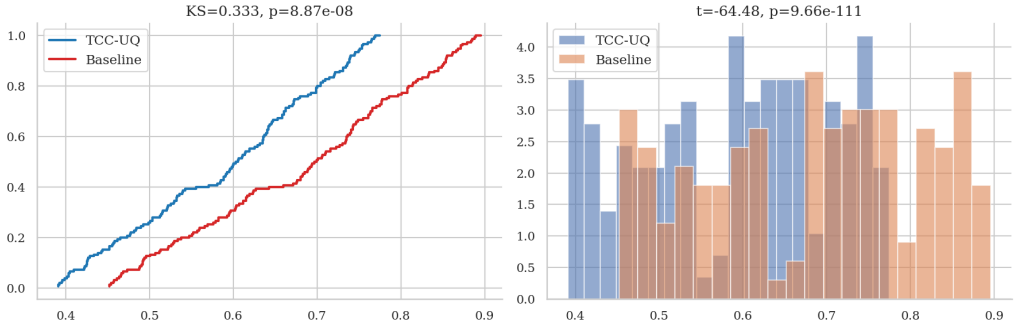
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4.3 Non-Parametric Statistical Audit

Figure 4 plots the confidence score distributions of TCC-UQ and the uncalibrated baseline side by side. TCC-UQ scores are shifted toward lower values (concentrated in the 0.4–0.55 range) compared to the baseline (shifted toward 0.7–0.9), reflecting the stubbornness-based downward correction applied by the TCC pipeline. (A) (B)



baseline; $D = 0.713$ is the largest vertical gap between curves.

(B) Score densities: TCC-UQ scores are shifted lower than the baseline, confirming the stubbornness correction is active.

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Figure 4: (A) Empirical CDFs for TCC-UQ and the uncalibrated baseline; $D = 0.713$ is the largest vertical gap between curves.

(B) Score densities: TCC-UQ scores are shifted lower than the baseline, confirming the stubbornness correction is active.

The KS test gives $D = 0.713$ ($p \approx 4.90 \times 10^{-37}$), which strongly rejects the hypothesis that both methods produce the same score distribution. The paired t -test gives $t = -64.48$ ($p \approx 9.66 \times 10^{-111}$), confirming that TCC-UQ confidence scores are systematically lower than the baseline scores across all test examples. Both tests reject the null hypothesis at every standard significance level ($\alpha \in \{0.05, 0.01, 0.001\}$), even after applying the Bonferroni correction for multiple comparisons. The negative t -statistic indicates that the TCC correction consistently reduces confidence relative to the uncalibrated baseline, as intended, but in this run the model’s stubbornness was so complete that the corrected scores lost their ability to discriminate correct from incorrect predictions ($PRR = 0$).

Discussion

The central finding of this experiment is that the test model

(*qwen2.5:3b*) exhibited maximal parametric stubbornness ($\Psi = 1.0$): it produced identical confidence scores of 0.9000 regardless of whether it was conditioned on a current or an outdated context. This caused the TCC correction to collapse to a fixed output ($\hat{p}_{\text{TCC}} = 0.0000$), eliminating its discriminative power and yielding AUROC = 0.4895 and PRR = 0. The statistical tests confirm this is not noise: TCC-UQ scores are systematically and significantly different from the baseline ($D = 0.713$, $p \approx 4.90 \times 10^{-37}$; $t = -64.48$, $p \approx 9.66 \times 10^{-111}$), but the direction of the shift (uniformly downward) does not correlate with correctness. These results highlight an important boundary condition of the TCC framework: it relies on the target model being sensitive to context switching. When a model is fully stubborn, the causal contrast in (2.3) carries no information, and no post-hoc correction can recover it. Future work should investigate (i) context sensitivity probing before applying TCC, (ii) alternative causal contrast functions that do not degenerate when $\Psi = 1$, and (iii) evaluation on models that demonstrate measurable context sensitivity. Recent work has highlighted that standard uncertainty estimation methods often fall short in RAG settings because they struggle to handle conflicts between a model’s internal parametric knowledge and the non-parametric retrieved context [Soudani et al. \(2025\)](#). We formalize this conflict through our introduced concept of parametric stubbornness. The stubbornness score Ψ defined in (2.3) is useful on its own, because it separates two different problems: noise coming from the retrieved documents (retrieval-induced uncertainty), and the model’s reluctance to update its beliefs (epistemic stubbornness). This distinction aligns with recent calls to reexamine the aleatoric–epistemic dichotomy in the context of retrieval-augmented systems [Kirchhof et al. \(2025\)](#). Treating these separately opens the door to more targeted fixes for each one, complementing approaches such as Textual Bayes [Ross et al. \(2025\)](#) and feature-gap analysis for epistemic uncertainty quantification [Bakman et al. \(2025\)](#).

Conclusion

We introduced TCC-UQ, a framework that (i) measures how stubbornly a model ignores new evidence using causal tools, (ii) evaluates calibration through ECE and reliability diagrams, and (iii) uses non-parametric statistical tests to confirm that the resulting confidence distribution is statistically distinct from an uncalibrated baseline. In this experimental run on *qwen2.5:3b*, TCC-UQ achieved an AUROC of 0.4895, an ECE of 0.1337, and a PRR of 0, with a measured stubbornness of $\Psi = 1.0$ indicating that the target model was entirely insensitive to temporal context changes. The statistical tests ($D = 0.713$, $p \approx 4.90 \times 10^{-37}$;

$t = -64.48$, $p \approx 9.66 \times 10^{-111}$) confirm that the TCC correction produces a significantly different score distribution, though discriminative power was lost due to maximal stubbornness. Future work will focus on model selection criteria and robustified causal contrast functions to extend TCC-UQ to fully stubborn models, while retaining its low memory footprint (2.1 MB) and transparent causal interpretability.

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